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What is the issue with the simple reduction? (Partition array, sum parts in parallel, sum partial results serially)

Does not scale to many processors

0%

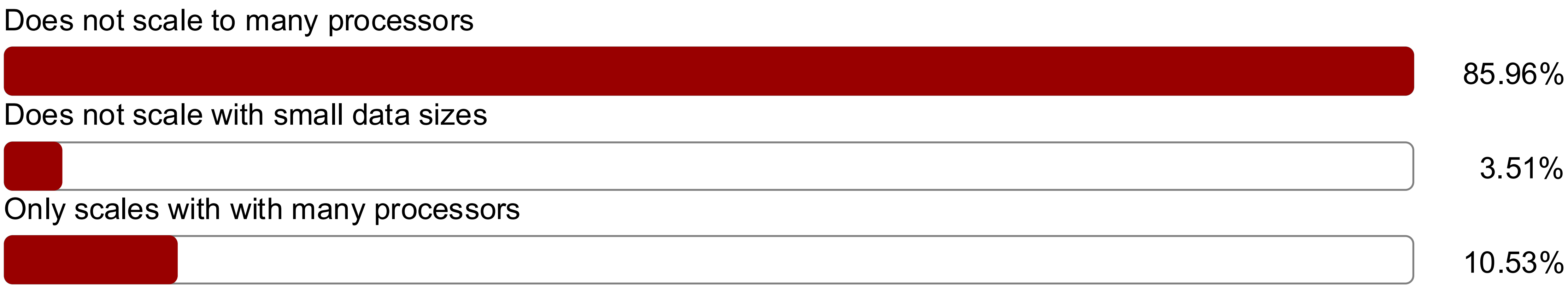
Does not scale with small data sizes

0%

Only scales with with many processors

0%

What is the issue with the simple reduction? (Partition array, sum parts in parallel, sum partial results serially)



What was the (theoretical max speed-up for the simple method given an array of length N?

N

0%

1/N

0%

sqrt(N)/2

0%

What was the (theoretical max speed-up for the simple method given an array of length N?

N	0%
1/N	1.54%
sqrt(N)/2 ✓	98.46%

RESULTS SLIDE

How many steps does the binary tree reduction need for an array of length N?

log2(N)

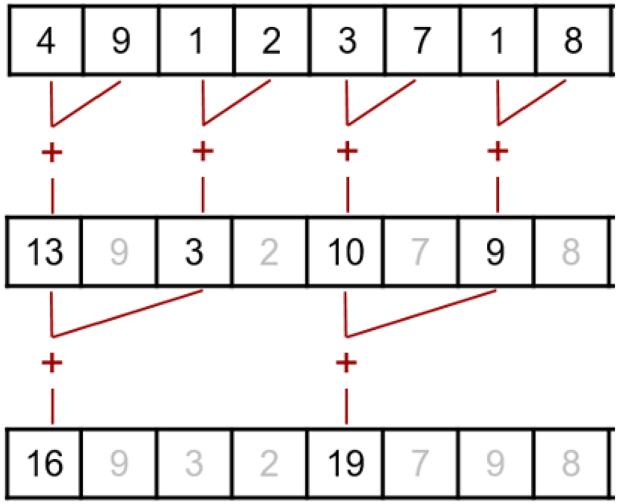
0%

N

0%

sqrt(N)

0%

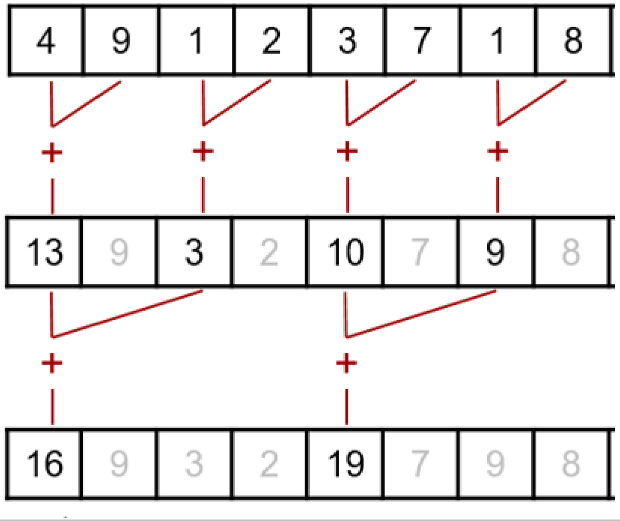


How many steps does the binary tree reduction need for an array of length N?

log2(N) ✓ — 98.59%

N — 1.41%

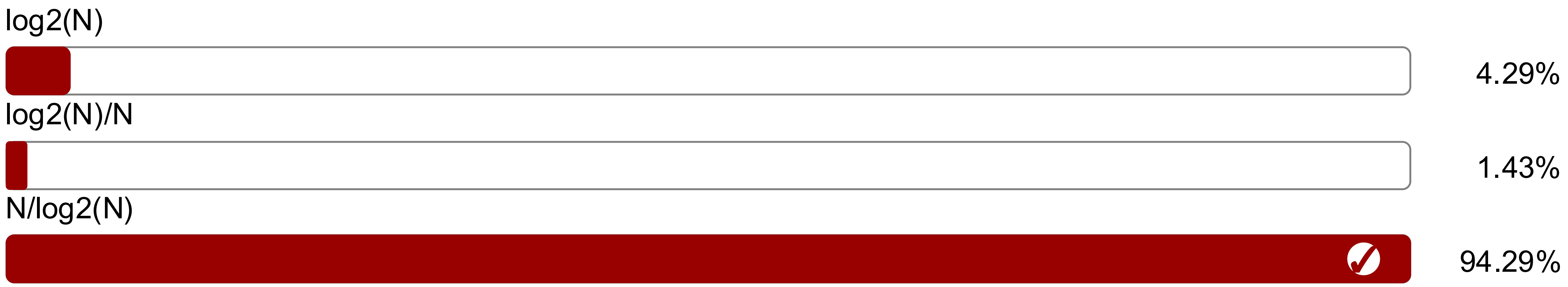
sqrt(N) — 0%



What was the (theoretical) max speed-up for the binary tree reduction given an array of length N?

Answer	Result
log2(N)	0%
log2(N)/N	0%
N/log2(N)	0%

What was the (theoretical) max speed-up for the binary tree reduction given an array of length N?



RESULTS SLIDE

Can multiply be a reduction operator?

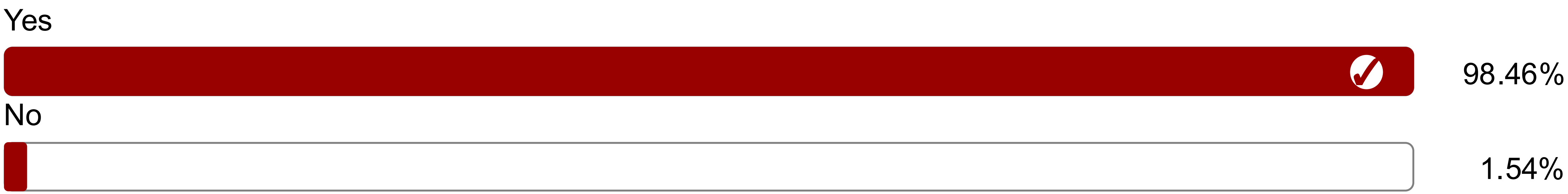
Yes

0%

No

0%

Can multiply be a reduction operator?



RESULTS SLIDE

Can subtract be a reduction operator?

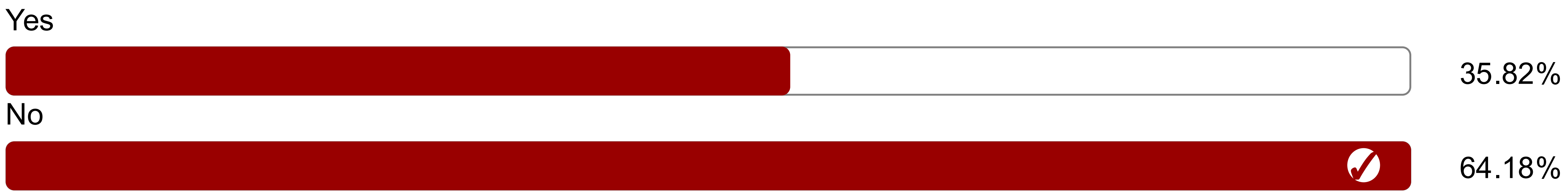
Yes

0%

No

0%

Can subtract be a reduction operator?



RESULTS SLIDE

Can max be a reduction operator?

Yes

0%

No

0%

Can max be a reduction operator?

Yes	100%
No	0%

RESULTS SLIDE

Can matrix multiplication be a reduction operator?

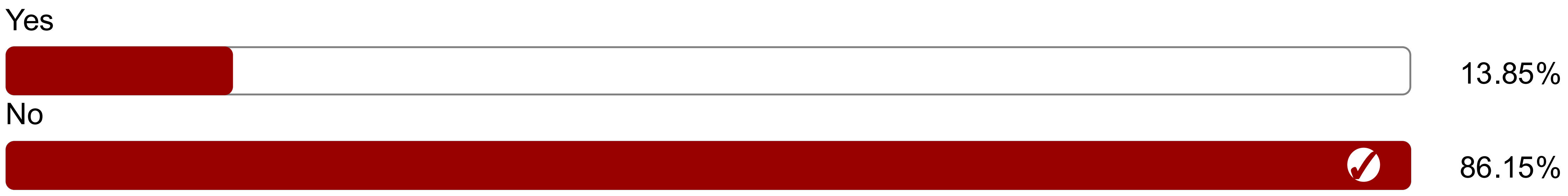
Yes

0%

No

0%

Can matrix multiplication be a reduction operator?



RESULTS SLIDE

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What is the principle of NUMA?

Memory cannot always be accessed

0%

Memory access times are not always equal

0%

Memory may be stored on disk

0%

What is the principle of NUMA?

Memory cannot always be accessed
0%

Memory access times are not always equal
100%

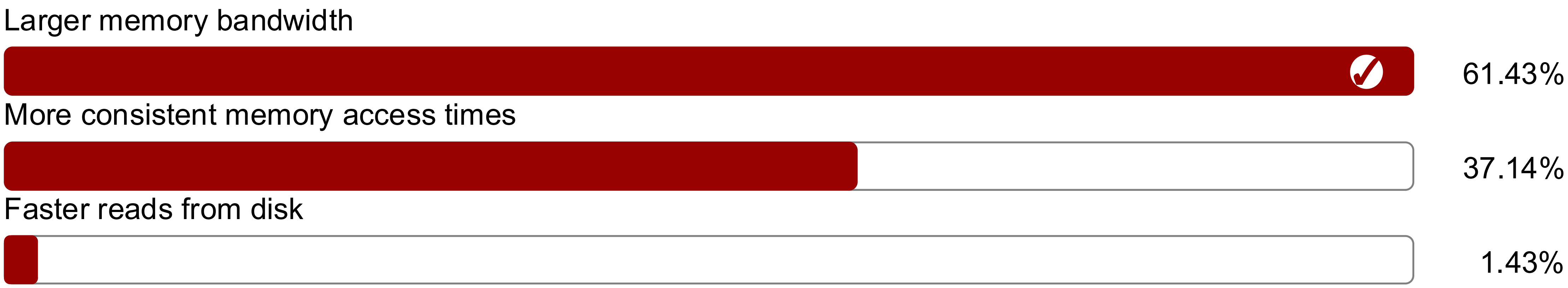
Memory may be stored on disk
0%

RESULTS SLIDE

Which of the following is a benefit of NUMA?

Larger memory bandwidth	0%
More consistent memory access times	0%
Faster reads from disk	0%

Which of the following is a benefit of NUMA?



RESULTS SLIDE

What was the purpose of numactl --interleave=all

To turn off NUMA	0%
To keep data on one memory bank	0%
To distribute data to all memory banks	0%

What was the purpose of numactl --interleave=all

To turn off NUMA

0%

To keep data on one memory bank

1.43%

To distribute data to all memory banks

98.57%

RESULTS SLIDE

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